Rotaract Club of Xiamen University Malaysia

Rotary Club of Kuala Lumpur Dynamics

Rotary Club of Tun Razak Exchange

SDC Hackathon

Technical Tools Declaration

Hosted by Xiamen University Malaysia (XMUM) AI Club

Submission Deadline: [Insert date]

Instructions: Declare every API, cloud service, AI tool, dataset, model, and sponsor resource your team used to build the project. Complete every section; where a category does not apply, tick "No" or write "None" – do not leave it blank. Submit as a single PDF document named "SDC2026_TeamName_TechDeclaration.pdf" together with your project submission. Undeclared third-party resources discovered at judging may affect your score or eligibility.

Instagram: @sdc_xmumhack ***Email:*** sdc.xmumhack@gmail.com

1. TEAM AND PROJECT OVERVIEW

Team Name

CTRL CV

Project / Solution Name

AI Workforce Pathon

Name (Leader)

MUHAMMAD ASYNAWI BIN AZWARRUDIN

Email & Phone (Leader)

asynawiazwarrudin@gmail.com

University Affiliation(s)

Universiti Teknikal Malaysia Melaka

Live Prototype URL

https://ctrlcv-ai-skill.replit.app/radar

Code Repository URL (public / access granted to judges)

https://github.com/AbdulMuhaimin949/Aiversecity-Future-Skills

2. ARCHITECTURE & TECHNOLOGY STACK

2.1. System Architecture Overview

Describe how the components declared in this document connect end-to-end (e.g., user interface → backend → AI model/API → database → cloud hosting). Attach a system block diagram as an appendix if available.

[Maximum 200 words]

AI Workforce Pathon is deployed as a modular, human-in-the-loop three-tier system:

1. User Interface: A responsive web dashboard developed in Python using Streamlit, deployed on Replit. It handles CV text/PDF ingestion, displays the evidence-backed skills review screen, scenario toggles, and radar visualizers.
2. Backend & Logic Layer: Built on Python. Ingests raw CV text, executes hybrid rule-based and semantic entity extraction, and evaluates profile embeddings against target role matrices across three disruption scenarios (Baseline, Moderate, Fast).
3. Cloud & Compute Infrastructure: Web hosting and user-facing sessions run continuously on Replit, while batch vector embedding indexing and trend-feed ingestion routines leverage containerized RunPod cloud compute instances.

2. ARCHITECTURE & TECHNOLOGY STACK

2.2. Languages, Frameworks & Libraries

List the main programming languages, frameworks, and third-party libraries that your team did not create. Open-source use is permitted provided it is declared and properly licensed/attributed.

Name & Version	Purpose in Project	License
Python 3.11	Core programming language for backend processing, the Medallion data pipeline (Bronze -> Silver -> Gold), and role-matching math.	PSF License
Streamlit 1.39+	Interactive web dashboard framework for CV ingestion, evidence-backed skill review, scenario toggles, and radar visualizers.	Apache-2.0
FastAPI 0.115+	Backend API layer orchestrating CV parsing, disruption scenario weighting, and vector similarity scoring.	MIT License
Pydantic 2.9+	Strict JSON schema validation and structured output enforcement for LLM responses.	MIT License
Pandas 2.2+ & NumPy 1.26+	Tabular data manipulation across pipeline datasets (synthetic_cvs.csv, cv_experiences.csv, role_match_explanations.csv)	BSD-3-Clause
Scikit-learn 1.5+	Mathematical computation of cosine similarity and normalized skill vector comparisons.	BSD-3-Clause

3. APIs & CLOUD SERVICES

3.1. External APIs

List every external API your solution calls (GenAI, maps, payments, messaging, data feeds, etc.). Do not include API keys or secrets in this document.

API and Provider	Purpose in Project	Access Tier (free / paid / sponsor-provided)
RunPod Serverless / Compute API — RunPod	Executes containerized batch vector embedding indexing and periodic labor-trend signal aggregation routines.	Sponsor-Provided
Google Gemini API (gemini-1.5-flash) — Google Cloud / Google AI Studio	Extracts structured skill entities and supporting CV evidence excerpts using Pydantic JSON schema constraints, and generates grounded role-match explanations	Free Tier (Google AI Studio)

3. APIs & CLOUD SERVICES

3.2. Cloud Platforms & Deployment

List every cloud service used for hosting, storage, databases, authentication, or compute, and indicate how the prototype is deployed.

Service & Provider	Purpose in Project	Plan / Region
Replit	Primary full-stack application hosting, development environment, and live judge demo URL	Sponsor-Provided
GitHub	Public source code repository, version control, and team collaboration	Free Public Tier
RunPod	Containerized cloud compute instances for batch vector embedding indexing and trend-feed ingestion routines.	Sponsor-Provided / Cloud Compute

4. AI TOOLS, MODELS & DATASETS

4.1. AI Models Used in the Solution

List every AI model embedded in or called by your application, including pre-trained, fine-tuned, and self-trained models.

Model & Version	Source / Provider	Usage (pre-trained / fine-tuned / self-trained)	License / Terms
Gemini 1.5 Flash	Google Cloud	Pre-trained (via API with Pydantic JSON schema constraints)	Google Generative AI API Terms
all-MiniLM-L6-v2 (Sentence-Transformers) / text-embedding-004	Hugging Face / Google Cloud	Pre-trained (used to generate normalized skill and role embeddings for cosine similarity matching)	Apache-2.0 / Google API Terms

4. AI TOOLS, MODELS & DATASETS

4.2. Datasets

List all datasets used for training, fine-tuning, evaluation, or in the running application. State clearly if a dataset contains personal or sensitive data.

Dataset Name	Source / URL	Type (public / private / self-collected / synthetic)	License	Personal Data? (Y/N)
synthetic_cvs.csv	Internal project repository (synthetic_cvs.csv)	Synthetic	MIT License (Internal Project Artifact)	N
cv_experiences.csv	Internal data pipeline (aiversecity/data/silver/cv_experiences.csv)	Synthetic (Processed)	MIT License (Internal Project Artifact)	N
role_match_explanations.csv	Internal data pipeline (aiversecity/data/gold/role_match_explanations.csv)	Synthetic (Processed)	MIT License (Internal Project Artifact)	N
World Industries & Skill Resilience Benchmarks	WEF Future of Jobs Report, Hays Asia Salary Guide 2025 & MOSTI Workforce Projections (world-industries.md, skill-resilience.md)	Public	Public Benchmark / Attribution	N
Verified Malaysian Reskilling & TVET Course Registry	Public catalogs from HRD Corp Malaysia and accredited TVET micro-credential providers	Public / Self-Collected Curated Registry	Public Domain / Attribution	N
UN SDG Goal 4 & Goal 8 Competency Mapping	United Nations Sustainable Development Goals Metadata (https://sdgs.un.org)	Public	Public Domain (UN)	N

4. AI TOOLS, MODELS & DATASETS

4.3. AI-Assisted Development Tools

Tools used to help write code, design, content, or documentation (e.g., chat assistants, code completion tools, image generators). Declaring these tools does not reduce your score; undeclared use may.

Did your team use any generative-AI tool to help create code, design, text, or other project material?

☑ Yes ☐ No

If yes, list them below:

Tool & Version	Used For	Extent of Use (minor / moderate / major)
Claude 5.5 Sonnet	Rapid code scaffolding, Streamlit component boilerplate.	Moderate
Google Gemini / ChatGPT	Technical documentation refinement, SDG impact mapping, and edge-case test data generation (synthetic_cvs.csv)	Minor
Claude Opus 5.5 (Anthropic)	End-to-end system architecture design from scratch, Medallion data pipeline (Bronze -> Silver -> Gold) logic, deterministic skill-resilience math, and complex debugging	Major

4. AI TOOLS, MODELS & DATASETS

Briefly describe how your team reviewed, tested, and took responsibility for AI-generated output:

[Maximum 100 words]

1. Development Code Governance: All AI-assisted code (via Replit Agent and LLM coding assistants) underwent mandatory line-by-line peer review, refactoring, and edge-case debugging before merging. We validated Medallion pipeline transformations (Bronze -> Silver -> Gold CSV outputs) and deterministic cosine-similarity math against known data quirks (pipeline-data-quirks.md).

2. Runtime Output Accountability: In-app Gemini outputs are constrained by strict Pydantic JSON schemas to prevent malformed responses. Crucially, our human-in-the-loop workflow requires users to verify and edit AI-extracted skills against verbatim CV evidence excerpts, while reskilling recommendations strictly query a verified course registry—marking unmapped gaps as "uncovered" to eliminate hallucination.

5. SPONSOR-PROVIDED RESOURCES

Declare any official SDC sponsor-provided GenAI API, cloud credits, tools, mentoring, or data that your team used.

Did your project use any official SDC sponsor-provided resource?

☑ Yes ☐ No

If yes, list them below:

Sponsor	Resource (API / credits / platform / data)	How It Was Used in the Project
Replit	Cloud IDE, full-stack hosting, and web deployment	Deployed the live Streamlit interactive prototype accessible to judges.
RunPod	Cloud compute credits & containerized GPU/CPU serverless instances	Executed containerized batch vector-embedding generation, cosine-similarity indexing, and periodic labor-trend aggregation workloads.
Google Cloud / Google AI (Include if provided via SDC sponsor credits/tier	Gemini API (gemini-1.5-flash & text-embedding-004)	Powered structured Pydantic JSON skill extraction from CV text and generated grounded role-match explanations (role_match_explanations.csv).

5. SPONSOR-PROVIDED RESOURCES

Was the sponsor resource essential to the core functionality of the prototype, or could it be replaced? Explain briefly:

[Maximum 100 words]

Sponsor resources were essential to our rapid prototyping velocity and live accessibility within hackathon timelines, while our core architecture remains fully portable and vendor-agnostic. Replit provided a zero-configuration cloud IDE, AI-assisted workspace, and instant public deployment for our full-stack web dashboard, while RunPod supported containerized batch vector-embedding and trend-indexing workloads. However, because the Medallion data pipeline (Bronze/Silver/Gold), deterministic cosine-similarity engine, and REST APIs are built on standard open-source Python/Node.js runtimes and environment-variable abstractions, the prototype can be seamlessly redeployed to any Docker-compatible cloud host with zero code refactoring.

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.1. Data Handling & Privacy

What user or personal data does your application collect, store, or send to third-party services (including AI APIs)? How is it protected, and how long is it kept? State "No user data is collected" if applicable.

[Maximum 150 words]

AI Workforce Pathon (Aiversecity Future Skills) strictly enforces privacy-by-design and Malaysian PDPA principles.
1. Collection & Pre-Transmission Redaction: When users upload a CV or paste career text, the file is parsed exclusively in volatile memory (RAM). Before any text is sent to third-party AI APIs (Google Gemini API), deterministic regex and local scrubbing strip Personally Identifiable Information (PII)—including names, NRIC/passport numbers, phone numbers, emails, and physical addresses.
2. Third-Party API Transmission & Protection: Only anonymized professional experience and skill text are transmitted over TLS 1.3/HTTPS to the Gemini API for structured skill extraction, while downstream role matching operates strictly on anonymized skill vectors. 3. Retention Policy: Zero persistent user data is stored in any database. All in-memory session states and temporary buffers are purged immediately upon session termination. Built-in demo workflows rely exclusively on synthetic profiles (synthetic_cvs.csv).

6. SECURITY, PRIVACY & RESPONSIBLE USE

6.2. Known Limitations & Risks

Identify the main technical limitations of your AI components (e.g., hallucination, bias, rate limits, dependency on a single provider) and any mitigation you have implemented.

[Maximum 150 words]

1. Generative Hallucination & Extraction Bias: LLMs may invent fictitious credentials or over-index on keyword-heavy CVs. Mitigation: Extraction is constrained by strict Pydantic JSON schemas requiring verbatim CV evidence excerpts, while human-in-the-loop review allows users to edit extracted skills before deterministic cosine-similarity scoring. Reskilling pathways query only verified HRD Corp and accredited registries; unverified gaps are explicitly flagged as "uncovered."

2. API Rate Limits & Single-Provider Dependency: Relying on external Gemini API endpoints introduces quota and latency risks during concurrent CV parsing. Mitigation: Pre-computed Gold-layer role embeddings (role_match_explanations.csv), deterministic local scoring, and response caching minimize live LLM calls, with modular SDK wrappers enabling seamless fallback to alternative models.

3. Labor Trend Uncertainty: Workforce projections are probabilistic estimates, not absolute truths. Mitigation: The engine models three disruption scenarios (Baseline, Moderate, Fast) and cites benchmark timestamps transparently.

7. COMPLIANCE & READINESS

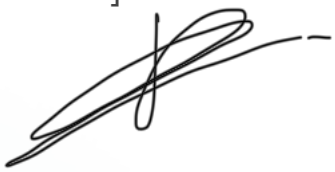
Tick all boxes that apply before submitting.

- [/] All APIs, cloud services, AI tools, models, and datasets used in the project are declared in this document.
- [/] All third-party code, datasets, and models are used in accordance with their licenses, and attribution is provided where required.
- [/] No API keys, passwords, or other credentials are exposed in the code repository, prototype, or this document.
- [/] Any use of generative-AI tools for development has been disclosed and reviewed by the team.
- [/] Any use of official SDC sponsor resources has been declared in Section 5.
- [/] The prototype described here is accessible to judges through the links provided in Section 1.
- [/] All fields in this template have been completed within the word limits stated.

8. DECLARATION

By submitting this document, the team leader confirms that the information provided is complete and accurate to the best of the team's knowledge, and that every API, cloud service, AI tool, dataset, model, and sponsor resource used in the project has been declared.

[Signature]



[Date]

25 October 2026